

Inverse Kazhdan–Lusztig polynomials of matroids need not be log-concave

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Abstract

We present an explicit simple binary matroid of rank 17 on 130,419 elements whose inverse Kazhdan–Lusztig polynomial has non-log-concave coefficients. This gives a counterexample to a conjecture of Gao and Xie. The matroid is obtained from a binary projective geometry by deleting a projective subspace and 589 further points in an affine subspace. We compute its inverse Kazhdan–Lusztig polynomial using a Gaussian-binomial recurrence and the deletion formula of Braden, Ferroni, Matherne, and Nepal.

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1 Introduction

The Kazhdan–Lusztig polynomial $P_M(t)$ of a matroid was introduced by Elias, Proudfoot, and Wakefield [EPW16]. Gao and Xie [GX21] introduced the inverse Kazhdan–Lusztig polynomial $Q_M(t)$ using the signed inverse of the Kazhdan–Lusztig function in the incidence algebra of the lattice of flats. Both polynomials have nonnegative integer coefficients by the singular Hodge theory of Braden, Huh, Matherne, Proudfoot, and Wang [BHMPW26].

A finite sequence $(a_0, \dots, a_d)$ is *log-concave* if $a_i^2 \geq a_{i-1}a_{i+1}$ for every $1 \leq i < d$. Gao and Xie [GX21, Conjecture 4.2] conjectured that the coefficients of $Q_M(t)$ are log-concave with no internal zeros for every matroid. The log-concavity assertion is also recorded as Conjecture 1.2(c) of Braden, Ferroni, Matherne, and Nepal [BFMN25]. We give an explicit counterexample. The construction is an explicit binary specialization of Cheng and Liu’s counterexamples to unimodality of ordinary Kazhdan–Lusztig polynomials [CL26].

Let $V = \mathbb{F}_2^{17}$ have basis $e_1, \dots, e_{17}$, and put

$$W = \text{span}(e_1, \dots, e_6), \quad H = \text{span}(e_7, \dots, e_{16}).$$

For $0 \leq m \leq 588$, write $m = \sum_{\ell=0}^9 b_\ell(m)2^\ell$ with $b_\ell(m) \in \{0, 1\}$, and define

$$v_m = e_{17} + \sum_{\ell=0}^9 b_\ell(m)e_{7+\ell}. \quad (1.1)$$

Let M be the binary matroid represented by the columns

$$E = V \setminus (W \cup \{v_0, \dots, v_{588}\}). \quad (1.2)$$

Thus M is obtained from $\text{PG}(16, 2)$ by deleting the 63 nonzero vectors of W and 589 points of the affine subspace $e_{17} + H$.

Theorem 1.1. *The matroid M defined by (1.1)–(1.2) is simple and has rank 17 and 130,419 elements. If $q_i = [t^i]Q_M(t)$, then*

$$q_2^2 < q_1 q_3.$$

In particular, inverse Kazhdan–Lusztig polynomials of loopless binary matroids need not have log-concave coefficients.

The proof has two steps. First, we compute the inverse Kazhdan–Lusztig polynomial of the complement of a projective subspace by a recurrence that groups flats according to their dimensions and intersections with the deleted subspace. Second, we apply the deletion formula of Braden, Ferroni, Matherne, and Nepal [BFMN25, Theorem 1.4]. The placement of the additional points ensures that each deletion changes only the constant and linear coefficients. The resulting linear coefficient is large enough to violate log-concavity at degree 2.

1.1 Related work

Gao and Xie [GX21] established the defining recursion and incidence-algebra identities for Q_M , proved multiplicativity, and computed its constant coefficient and its values on Boolean and uniform matroids. They formulated both nonnegativity and log-concavity conjectures. Proudfoot [Pro21] placed the inverse polynomials in the equivariant Kazhdan–Lusztig–Stanley framework. Braden, Huh, Matherne, Proudfoot, and Wang [BHMPW26, Theorem 1.5] proved nonnegativity, including an equivariant version. The present example concerns the stronger inequalities between consecutive coefficients.

Several families satisfy these inequalities. Gao and Xie [GX21, Proposition 4.3] proved log-concavity for uniform matroids in their original paper. Gao, Xie, and Yang [GXY22] computed equivariant inverse Kazhdan–Lusztig polynomials of uniform matroids. Gao, Li, Xie, Yang, and Zhang [GLXYZ26] proved induced log-concavity results for uniform and q -niform matroids, which imply ordinary log-concavity. Xie and Zhang [XZ26] proved log-concavity for paving matroids. Gao, Li, and Xie [GLX25] established log-concavity for thagomizer matroids and the graphic matroids of $K_{2,n}$, and Gao, Li, and Li [GLL26] obtained formulas and log-concavity for fan matroids. Xie, Zhang, and Zhong [XZZ26] obtained further equivariant inverse formulas for thagomizer matroids under the hyperoctahedral group action.

Other work supplies structural tools and explicit formulas. Vecchi [Vec21] studied modularity and the leading coefficient of the inverse polynomial. Gao, Proudfoot, Yang, and Zhang [GPYZ24] computed leading inverse coefficients for braid matroids. Ferroni and Vecchi [FV22] analyzed circuit-hyperplane relaxation. Ferroni, Nasr, and Vecchi [FNV23] developed stressed-hyperplane relaxation and formulas for paving matroids, and Karn, Nasr, Proudfoot, and Vecchi [KNPV23]

extended these methods to equivariant invariants. Ardila and Sanchez [AS23] proved valuativity of the inverse polynomial, and Ferroni and Schröter [FS24] developed valuative formulas for large classes of matroids.

Braden and Vysogorets [BV20] proved deletion formulas for ordinary Kazhdan–Lusztig and Z -polynomials, the latter introduced by Proudfoot, Xu, and Young [PXY18]. Braden, Ferroni, Matherne, and Nepal [BFMN25] established the corresponding inverse formulas and applied them to uniform matroids, projective geometries, glued cycles, and matroids of corank two. They also constructed a rank-19 counterexample to a real-rootedness conjecture of Xie and Zhang about a binomially normalized inverse Kazhdan–Lusztig polynomial [BFMN25, Theorem 1.5]. Their counterexample addresses that real-rootedness assertion; Theorem 1.1 gives a failure of log-concavity for the coefficients of Q_M itself.

Our construction starts with a Bose–Burton geometry, obtained by removing a projective subspace from a projective geometry [BB66]. Our choice of additional deleted points specializes the construction of Cheng and Liu [CL26], who show that ordinary Kazhdan–Lusztig polynomials need not be unimodal. Their construction produces examples over every finite field and includes binary examples of rank 17 on 130,419 elements. Here we compute the inverse polynomial of the specified binary example and verify the distinct coefficient inequality in Theorem 1.1.

Section 2 fixes conventions and recalls the two inverse Kazhdan–Lusztig identities used in the proof. Section 3 derives the recurrence for complements of projective subspaces. Section 4 controls the effect of the additional deletions, and Section 5 gives the exact computation proving Theorem 1.1.

1.2 Declaration of generative AI usage

The counterexample was generated autonomously by Rethlas, the AI agent for mathematical research described by Ju et al. [Ju+26], using GPT-5.6 Sol Max as its base model. AI assistance was also used to prepare and edit this manuscript. Ronnie Cheng and Shurui Liu verified the proof.

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2 Preliminaries

All matroids are finite and loopless. We write $L(N)$ for the lattice of flats of N , rk_N for its rank function, $N|F$ for restriction to a flat F , and N/F for contraction. Both P_N and Q_N depend only on the ranked lattice of flats. In particular, simplification does not change either polynomial.

We use the palindromic characterization of P_N following Proudfoot, Xu, and Young [PXY18], as stated by Braden and Vysogorets [BV20, Theorem 2.2]: $P_N(t) = 1$ when $\text{rk } N = 0$, $\deg P_N(t) < \text{rk } N/2$ when $\text{rk } N > 0$, and

$$Z_N(t) = \sum_{F \in L(N)} t^{\text{rk}_N F} P_{N/F}(t) \quad (2.1)$$

satisfies $t^{\text{rk } N} Z_N(t^{-1}) = Z_N(t)$.

For a matroid K , set

$$\tau(K) = \begin{cases} [t^{(\text{rk } K - 1)/2}] P_K(t), & \text{rk } K \text{ odd,} \\ 0, & \text{rk } K \text{ even.} \end{cases}$$

We write

$$\begin{bmatrix} a \\ b \end{bmatrix}_2 = \prod_{u=0}^{b-1} \frac{2^{a-u} - 1}{2^{b-u} - 1} \quad (0 \leq b \leq a)$$

for the binary Gaussian binomial coefficient, and set it equal to zero when $b < 0$ or $b > a$. It counts the b -dimensional subspaces of $\mathbb{F}_2^a$.

Proposition 2.1 (Gao–Xie [GX21]). *If N has positive rank and ground set E_N , then*

$$Q_N(t) = - \sum_{\substack{F \in L(N) \\ F \neq E_N}} Q_{N|F}(t) (-1)^{\text{rk}(N/F)} P_{N/F}(t). \quad (2.2)$$

The rank-zero initial value is $Q_N(t) = 1$.

This is the fourth identity of Gao–Xie [GX21, Theorem 1.3]. Equivalently, on each interval $[F, G]$ of $L(N)$, the function $(-1)^{\text{rk } G - \text{rk } F} Q_{(N|G)/F}(t)$ is the incidence-algebra inverse of the function $P_{(N|G)/F}(t)$.

Theorem 2.2 (Braden–Ferroni–Matherne–Nepal [BFMN25]). *Let N be a simple matroid, and let i be an element that is not a coloop. Put*

$$\mathcal{T}_i(N) = \{F \in L(N) : i \in F, F \setminus \{i\} \notin L(N)\}.$$

Then

$$Q_N(t) = Q_{N \setminus i}(t) + (1+t) Q_{N/i}(t) - \sum_{\substack{F \in \mathcal{T}_i(N) \\ \text{rk } F \text{ even}}} \tau((N|F)/i) t^{\text{rk } F/2} Q_{N/F}(t). \quad (2.3)$$

Equation (2.3) is the inverse Kazhdan–Lusztig part of [BFMN25, Theorem 1.4]. Since i is not a loop, $\text{rk}((N|F)/i) = \text{rk } F - 1$. Thus the summands with odd $\text{rk } F$ vanish by the definition of τ ; restricting the sum to even ranks makes all displayed exponents integral.

Lemma 2.3. *Let $G_a = \text{PG}(a-1, 2)$ for $a \geq 1$, and let G_0 be the empty rank-zero matroid. Then*

$$P_{G_a}(t) = 1, \quad Q_{G_a}(t) = 2^{\binom{a}{2}}. \quad (2.4)$$

Proof. The lattice of flats of G_a is the lattice of subspaces of $\mathbb{F}_2^a$, and every contraction by a flat simplifies to a projective geometry. Induction in (2.1) shows that $P_{G_a} = 1$: with this choice,

$$Z_{G_a}(t) = \sum_{d=0}^a \begin{bmatrix} a \\ d \end{bmatrix}_2 t^d,$$

which is palindromic because $\begin{bmatrix} a \\ d \end{bmatrix}_2 = \begin{bmatrix} a \\ a-d \end{bmatrix}_2$. The degree condition holds as well. This is the projective case of the modularity criterion in [EPW16, Proposition 2.14].

The ordinary Kazhdan–Lusztig polynomial on every interval is therefore 1, so its incidence-algebra inverse is the Möbius function of the subspace lattice. The Gaussian-binomial identity

$$\sum_{d=0}^a (-1)^d 2^{\binom{d}{2}} \begin{bmatrix} a \\ d \end{bmatrix}_2 = 0 \quad (a > 0)$$

gives $\mu(0, \mathbb{F}_2^a) = (-1)^a 2^{\binom{a}{2}}$ by induction. The sign convention in Proposition 2.1 yields $Q_{G_a} = 2^{\binom{a}{2}}$. The constancy of Q for modular matroids is also part of Vecchi’s characterization [Vec21, Theorem 3.5]. $\square$

3 Complements of projective subspaces

For $0 \leq k < r$, let $B_{r,k}$ be the rank- r binary matroid represented by the vectors of $\mathbb{F}_2^r \setminus W$, where W is a fixed k -dimensional subspace. Its isomorphism type depends only on r and k . In particular, $B_{r,0} = G_r$.

Cheng and Liu [CL26, Remark 2.7] computed the ordinary polynomial (3.2) and the corresponding identity (3.4). We recall their subspace argument and use it to derive the inverse recurrence (3.3).

Proposition 3.1. *For $1 \leq d \leq r-1$ and $0 \leq j \leq \min(k, d-1)$, put*

$$N_{r,k}(d, j) = 2^{(k-j)(d-j)} \begin{bmatrix} k \\ j \end{bmatrix}_2 \begin{bmatrix} r-k \\ d-j \end{bmatrix}_2. \quad (3.1)$$

Then

$$P_{B_{r,k}}(t) = \sum_{0 \leq u < r/2} \left(\begin{bmatrix} k \\ u \end{bmatrix}_2 - \begin{bmatrix} k \\ r-u \end{bmatrix}_2 \right) t^u, \quad (3.2)$$

and

$$\begin{aligned} Q_{B_{r,k}}(t) &= (-1)^{r+1} P_{B_{r,k}}(t) \\ &\quad + \sum_{d=1}^{r-1} \sum_{j=0}^{\min(k, d-1)} (-1)^{r-d+1} N_{r,k}(d, j) Q_{B_{d,j}}(t). \end{aligned} \quad (3.3)$$

Together with $Q_{B_{1,0}} = 1$, this is a finite recurrence for all $Q_{B_{r,k}}$.

Proof. A nonempty flat of $B_{r,k}$ is the set $U \setminus W$ for a subspace $U \not\subseteq W$. These surviving vectors span U : if $u \in U \setminus W$ and $w \in U \cap W$, then u and $u+w$ both survive and have sum w . Writing $d = \dim U$ and $j = \dim(U \cap W) < d$, the restriction to this flat is isomorphic to $B_{d,j}$.

To count the subspaces of this type, first choose $J = U \cap W$ and the image A of U in $\mathbb{F}_2^r/W$. There are $\begin{bmatrix} k \\ j \end{bmatrix}_2$ and $\begin{bmatrix} r-k \\ d-j \end{bmatrix}_2$ choices, respectively. Once these are fixed, the possible subspaces U/J are graphs of linear maps $A \rightarrow W/J$, giving $2^{(k-j)(d-j)}$ choices. This proves (3.1).

Every contraction by a nonempty flat simplifies to a projective geometry. Indeed, if the flat spans $U \not\subseteq W$, a coset $v+U$ representing a nonzero quotient vector cannot be wholly contained in W : otherwise differences of its vectors would give $U \subseteq W$. Thus every nonzero quotient direction has a surviving representative. Lemma 2.3 gives $P_{B_{r,k}/F} = 1$ for every nonempty flat F .

Put $\mathcal{G}_a(t) = \sum_{u=0}^a \begin{bmatrix} a \\ u \end{bmatrix}_2 t^u$. The nonempty flats correspond to precisely the subspaces not contained in W , so

$$Z_{B_{r,k}}(t) = P_{B_{r,k}}(t) + \mathcal{G}_r(t) - \mathcal{G}_k(t). \quad (3.4)$$

For $0 \leq u < r/2$, compare the coefficients of t^u and t^{r-u} in this palindromic polynomial. The degree bound on P gives

$$[t^u]P_{B_{r,k}} = \begin{bmatrix} r \\ r-u \end{bmatrix}_2 - \begin{bmatrix} k \\ r-u \end{bmatrix}_2 - \begin{bmatrix} r \\ u \end{bmatrix}_2 + \begin{bmatrix} k \\ u \end{bmatrix}_2 = \begin{bmatrix} k \\ u \end{bmatrix}_2 - \begin{bmatrix} k \\ r-u \end{bmatrix}_2,$$

proving (3.2).

Finally, apply (2.2). The empty flat contributes $(-1)^{r+1}P_{B_{r,k}}$. Each proper nonempty flat of type (d, j) contributes $(-1)^{r-d+1}Q_{B_{d,j}}$, and (3.1) gives its multiplicity. This proves (3.3). $\square$

For the parameters used in Theorem 1.1, (3.2) gives

$$P_{B_{17,6}}(t) = 1 + 63t + 651t^2 + 1395t^3 + 651t^4 + 63t^5 + t^6. \quad (3.5)$$

Evaluating the integer recurrence (3.3) gives

$$\begin{aligned} Q_{B_{17,6}}(t) = & 87028570972269444939528239473208013619200 \\ & + 83686001274920020390933101403804008448t \\ & + 8796350004117854247909387011096576t^2 \\ & + 164253785728427963887548104704t^3 \\ & + 622091629393725177724928t^4 \\ & + 468758076371402752t^5 + 55033842688t^6. \end{aligned} \quad (3.6)$$

For completeness, the double sum in (3.3), in ascending coefficient order, is

$$\begin{aligned} & (87028570972269444939528239473208013619199, \\ & 83686001274920020390933101403804008385, \\ & 8796350004117854247909387011095925, \\ & 164253785728427963887548103309, \\ & 622091629393725177724277, \\ & 468758076371402689, 55033842687). \end{aligned}$$

Adding the coefficients of (3.5) produces (3.6). Thus the computation uses only the finite Gaussian-binomial recurrence.

4 Additional point deletions

We now identify a condition under which additional point deletions change only the constant and linear coefficients of the inverse Kazhdan–Lusztig polynomial. The case of a single deletion from a full projective geometry was computed by Braden, Ferroni, Matherne, and Nepal [BFMN25, Proposition 4.4]; the following argument allows ordered deletions after a projective subspace has been removed.

Proposition 4.1. *Let $r \geq 2$, let $W \leq \mathbb{F}_2^r$ have dimension $k < r$, and let $p_1, \dots, p_s$ be distinct vectors outside W , where $s \geq 0$. For $0 \leq j \leq s$, set*

$$D_j = (W \setminus \{0\}) \cup \{p_1, \dots, p_j\},$$

and let M_j be represented by the nonzero vectors outside D_j . Suppose that every M_j has rank r , and that, for each $1 \leq j \leq s$, no pair $\{x, x + p_j\}$ is contained in D_{j-1} . Then

$$Q_{M_s}(t) = Q_{B_{r,k}}(t) - s2^{\binom{r-1}{2}}(1+t) + 2^{\binom{r-2}{2}} \left(s(2^{r-1} - 2^k) - \binom{s}{2} \right) t. \quad (4.1)$$

Proof. Fix j , and set $N = M_{j-1}$ and $i = p_j$. The rank assumption implies that i is not a coloop, so Theorem 2.2 applies. Each nonzero vector in $\mathbb{F}_2^r / \langle i \rangle$ has a two-element fiber $\{x, x + i\}$. By hypothesis, at least one of these vectors survives. Hence N/i simplifies to G_{r-1} , and

$$Q_{N/i}(t) = 2^{\binom{r-1}{2}}. \quad (4.2)$$

More generally, let $F \in \mathcal{T}_i(N)$ and let U be its vector span. Since F is a flat, every surviving vector in U belongs to F . Each nonzero vector of $U/\langle i \rangle$ has the same two-element fiber, with at least one surviving vector. Thus $(N|F)/i$ simplifies to $G_{\text{rk } F-1}$ and has ordinary Kazhdan–Lusztig polynomial 1. Consequently,

$$\tau((N|F)/i) = \begin{cases} 1, & \text{rk } F = 2, \\ 0, & \text{rk } F \neq 2. \end{cases} \quad (4.3)$$

A rank-two flat through i belongs to $\mathcal{T}_i(N)$ exactly when the other two points of its binary projective line both survive. Indeed, these two points span i ; if just one survives, it is a singleton flat of the simple matroid N . There are $2^{r-1} - 1$ projective lines through i . Each point of D_{j-1} excludes one such line, and distinct deleted points exclude distinct lines: a collision would give a wholly deleted pair $\{x, x + i\}$. The number of contributing rank-two flats is therefore

$$(2^{r-1} - 1) - ((2^k - 1) + (j - 1)) = 2^{r-1} - 2^k - j + 1. \quad (4.4)$$

For each contributing flat F , a coset modulo its two-dimensional span is a union of two pairs differing by i . Each pair contains a surviving vector, so every nonzero quotient direction is represented. Thus N/F simplifies to G_{r-2} and

$$Q_{N/F}(t) = 2^{\binom{r-2}{2}}. \quad (4.5)$$

Rearranging (2.3) and substituting (4.2)–(4.5) gives

$$Q_{M_j}(t) = Q_{M_{j-1}}(t) - 2^{\binom{r-1}{2}}(1+t) + (2^{r-1} - 2^k - j + 1)2^{\binom{r-2}{2}}t.$$

Sum over $1 \leq j \leq s$ and use $M_0 = B_{r,k}$ to obtain (4.1). $\square$

5 The explicit counterexample

We verify that the matroid in the introduction meets the hypotheses of Proposition 4.1, and then compute its inverse Kazhdan–Lusztig polynomial.

Lemma 5.1. *The matroid defined by (1.1)–(1.2) is simple of rank 17 on 130,419 elements. The ordered deletions $p_j = v_{j-1}$, $1 \leq j \leq 589$, satisfy the hypotheses of Proposition 4.1.*

Proof. The 589 vectors v_m are distinct points of $e_{17} + H$, which is disjoint from W . The deleted set in $\text{PG}(16, 2)$ has size $(2^6 - 1) + 589 = 652$, leaving

$$2^{17} - 1 - 652 = 130,419$$

elements. All representing columns are nonzero and distinct, so the binary matroid is simple.

The vectors

$$e_7, \dots, e_{16}, \quad e_i + e_7 \ (1 \leq i \leq 6), \quad e_{17} + e_1$$

all survive and span V : the first group supplies e_7 , the second then supplies $e_1, \dots, e_6$, and the last supplies e_{17} . They also survive every intermediate deletion, so all intermediate matroids have rank 17.

It remains to check the pair condition. Fix $p = v_m \in e_{17} + H$. The sum of two vectors in W belongs to W , and the sum of two vectors in $e_{17} + H$ belongs to H , so neither can equal p . The sum of a nonzero vector of W and a vector in $e_{17} + H$ has a nonzero W -component and therefore cannot equal p either. Thus no two previously deleted vectors sum to p , which is equivalent to the required pair condition. $\square$

Proof of Theorem 1.1. By Lemma 5.1, we may use (4.1) with $r = 17$, $k = 6$, and $s = 589$. The accumulated number of rank-two correction terms is

$$589(2^{16} - 2^6) - \binom{589}{2} = 589 \cdot 65472 - 173166 = 38389842.$$

Consequently,

$$Q_M(t) = Q_{B_{17,6}}(t) - 589 \cdot 2^{120}(1 + t) + 38389842 \cdot 2^{105}t. \quad (5.1)$$

The two changes in coefficients are

$$\begin{aligned} 589 \cdot 2^{120} &= 782915289517315449140342358505122955264, \\ -589 \cdot 2^{120} + 38389842 \cdot 2^{105} &= 774361710609625052082473627903779143680. \end{aligned}$$

Combining (3.6) and (5.1) gives

$$\begin{aligned} Q_M(t) &= 86245655682752129490387897114702890663936 \\ &\quad + 858047711884545072473406729307583152128t \\ &\quad + 8796350004117854247909387011096576t^2 \\ &\quad + 164253785728427963887548104704t^3 \\ &\quad + 622091629393725177724928t^4 \\ &\quad + 468758076371402752t^5 + 55033842688t^6. \end{aligned} \quad (5.2)$$

Let $q_i = [t^i]Q_M(t)$. Exact multiplication yields

$$\begin{aligned} q_2^2 &= 77375773394944174444152196477144115936648327953467708040957998923776, \\ q_1q_3 &= 140937585012651958843285045885463832582304464476783152190088504410112. \end{aligned}$$

Equivalently,

$$\begin{aligned} q_2^2 - q_1q_3 &= \\ -63561811617707784399132849408319716645656136523315444149130505486336 &< 0. \end{aligned} \quad (5.3)$$

This violates log-concavity at degree 2 and proves the theorem. $\square$

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